Integrative Multi-Omics Approach to LiCINS-F1 Regulatory Specificity in Lavender Glandular Trichomes

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Questions:

1. How many multi-omics integration layers do I need to select an element as a prioritized candidate for wet-lab validation?
2. Are these integrated omics-layer data enough to provide sufficient evidence for reporting a candidate, or would you suggest adding other analyses and datasets? Do you think adding an ATAC-seq layer could be useful?
3. Also, for controlling the complexity of these regulatory interactions, what would you recommend, or do you think this approach and stepwise analysis would be efficient for finding significant associations?
4. And in conclusion, I want to know how feasible this approach is and whether it is a logical design for studying this problem and providing useful regulatory information as a basis for precise bioengineering approaches in lavender.

A recent study in 2026 on trichome-specific promoters, it was demonstrated that **the LiCINS-F1 promoter drives GFP expression specifically in glandular trichome (GT) cells, but F2 to F4 showed non-specific expression (1).** This cell type is an important target for advanced bio-engineering in lavender to improve and increase terpene production. (2,3).

Why is this finding important?

For providing a more specific target for further research in this field, such as obtaining more precise information on bioengineering targets and achieving more efficient results. However, this research needs more exploration of the regulatory aspect of these cis-regulatory elements and their function. Trichome-specific promoter studies in other species have shown that promoter specificity can depend on combinations of activating and repressing cis-regions (4) (5). This study showed the foundation for next-step questions, such as **which specific molecular elements and regulatory mechanisms support specificity.** Additionally, Regulatory motifs could be involved in loss of specificity. Answering these details of this process would enhance bioengineering in lavender.

Why do we need a multi-omics approach?

According to the availability of the chromosome-level lavender genome and transcriptomic datasets and previously reported TF information, we can construct coexpression-based terpenoid-gene networks **to better analyze these regulatory aspects of the promoter, investigate, and interpret these biomolecular interactions (6–8).** Exploring each layer of omics data and integrating these available data on cis-element interactions would **give a more mechanistically informative view of specificity in the promoter in these GT cells.** Plant regulatory resources such as PlantRegMap can also provide more information on candidate TF-binding sites. (9). This is important for **prioritising a few best candidates for further wet-lab validation** to make this research process also **more efficient and quicker.**

How will the data be integrated? To elaborate on this, we can **find the motif segments in F1 and compare them with other motifs and corresponding regions in F2–F4** based on specificity and strength, and find the motifs associated with this specificity, and vice versa. This approach is supported by previous glandular-trichome promoter studies in which promoter deletion analysis was used to localize regions responsible for tissue specificity. (11). Additionally, a certain combination of motifs could be responsible for specificity rather than a single motif, such as MYB-binding motifs (11)(4). The previous Y1H study by Sarker et al. reported related candidate group TFs from MYB, bZIP, NAC, and GeBP. Exploring these complementary layers and integrating them with promoter sequence analysis could provide a more mechanistic view of the regulatory basis of promoter specificity in GTs. It should explore the previously reported TFs, such as MYB, bZIP, NAC, GeBP, and SBP-related factors identified in the lavender LiCINS/LiLINS Y1H study, as well as other biologically relevant TF families such as bHLH and HD-ZIP that have been involved in glandular-trichome regulation in other lavender or plant systems. (10,12–14). Trichome-specific promoters in the menthol promoter study have shown that fine-mapping through promoter deletion analysis can identify regions responsible for specificity and can help identify candidate cis-elements and TF-binding motifs (11). Also, exploring whether **certain TFs that interact with specific motifs are expressed or enriched in GT cells** is important. **Coexpression analysis approaches such as WGCNA could** also help identify trichome-associated genes and gene modules linked to terpene biosynthesis, provided that suitable GT-enriched or tissue-matched expression datasets are available(15). These multiple layers provide more convincing evidence to select our best candidates, while relying on only one layer may not be enough. In addition to this, **we can add an ATAC-seq analysis layer to assess chromatin accessibility**. ATAC-seq has been used to compare chromatin accessibility between glandular trichomes and other tissues in Artemisia annua and has shown the association of chromatin accessibility with certain motifs and tfs binding. (16). Finally, **the selected candidates that have passed a predefined evidence-score threshold** can be passed on for **wet-lab validation to test the functional and causal contribution, including reporter assays, motif mutation, EMSA/Y1H, and ChIP-qPCR**. These approaches can provide complementary evidence for promoter activity, motif requirement, DNA–protein interaction, and TF binding in a cellular context. Assays, promoter deletion/fine-mapping, and Y1H/LUC-type assays were useful in trichome promoter studies. (10,11) (14).

What is the final goal? **I want to integrate data from promoter sequence analysis, TF data, RNA-seq, coexpression, ATAC-seq data, and comparative genomics analysis to identify and prioritize the regulatory elements as final candidates for experimental evaluation.** The chromosome-level lavender genome and comparative sequence data se are very useful sources for studying promoter and regulatory analysis. (7). The final goal is to select a **few high-priority TF–cis-element interactions** for experimental wet lab evaluation and for further LiCINS-F1 promoter engineering. Promoter activity depends on many factors, including motif identity, position, spacing, orientation, copy number, and motif combinations. (17–19). Therefore, the computational analysis not only finds candidate regulatory elements but also provides the molecular basis information for designing targeted LiCINS-F1 variants for the next experimental and bioengineering implementation.

What are the main challenges? There are several main challenges in this approach. Firstly, the correlation data of motifs with a TF and GT-specific expression data does **not necessarily demonstrate a causal regulatory interaction.** Secondly, **the specificity might depend on a combination of several motifs, not only one element,** and this can make the interpretation even more complex. (10,11) (4). Additionally, there is a problem with the availability and quality of GT-specific omics data. This means that RNA-seq or coexpression data mainly provide association and expression information, and ATAC-seq indicates chromatin accessibility, but they **cannot represent direct motif function or TF binding**. (15) (16).

Therefore, to better assess this phenomenon and provide necessary information for bioengineering the promoters, we need to use integration of multidisciplinary approaches: computational, molecular, and functional evidence.

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